edge wrap

T2 Two-sided tests

Print pages 1 and 2 on the two sides of one card, flipping on the long edge.

H1 BUS A · NC

R1

R2

2 Vias

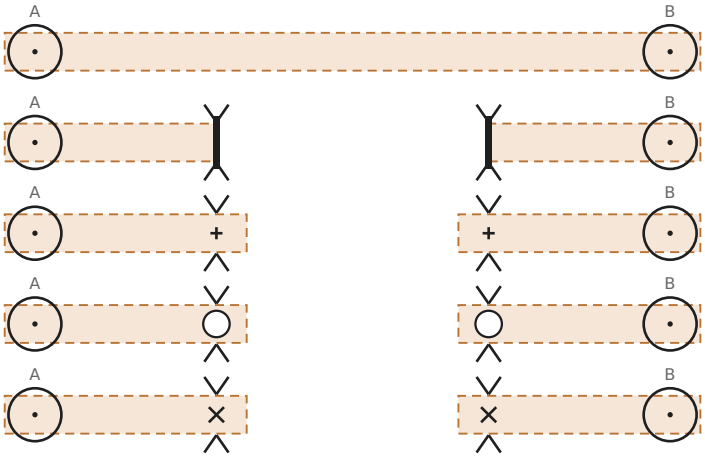
V0 Front only
the reference

V1 Slit vias
tape threads through

V2 Wire vias
leg through, soldered

V3 Eyelet vias
eyelet through both

V4 Your version



H2 V+ · NC

1 Registration

Push a pin through each corner mark, R1 to R4, and read the offset on the back.

3 Edge wraps

Fold 12 mm tape over the top edge, down to the dashed line on both sides.

4 Stacking shorts

The 12 mm front pad ends at S2. Behind it, a back pad joins S1 through a wire via. Sheet 3 says how to stack two sheets.

H3 GND · NC

H4 BUS B · NC

R3

R4

Print at 100 %: this bar should measure 100 mm.

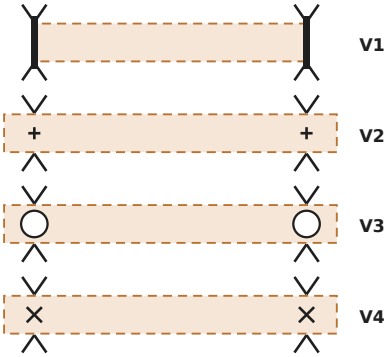
T2 Two-sided tests: back

This side prints on the back of page 1. Each mark should sit behind its partner.

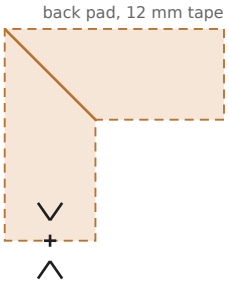
H1 BUS A · NC

R2

R1



H2 V+ · NC



H3 GND · NC

H4 BUS B · NC

R4

R3

Print at 100 %: this bar should measure 100 mm.

T2 Running the two-sided tests

Print this sheet once on plain paper. Print pages 1 and 2 double-sided on card, four copies.

H1 BUS A · NC

You need

- Four cards printed double-sided (pages 1 and 2)
- 5 mm and 12 mm copper tape, a craft knife and a pin
- Cut-off resistor legs and solder; small eyelets, about 3 mm
- A binder clip, a sheet of plain paper and one of vellum
- A multimeter with a 200 Ω range

1 · Registration

1. Check the 100 mm bar on both sides of one card.
2. Push a pin straight through the centre of each front mark, R1 to R4.
3. On the back, read where each hole landed on the 1 mm scale, across and up or down.

	Across	Up / down
R1		
R2		
R3		
R4		

H2 V+ · NC

Pass: within 1.5 mm everywhere. Lanes overlap 4 mm at wire and eyelet vias.

2 · Vias

1. **V1:** cut the two slits. Lay one piece of tape from A, through the first slit, along the back and up through the second slit, to B.
2. **V2:** lay the three pieces. Push a resistor leg through each via mark, bend it flat on both sides and solder it.
3. **V3:** lay the three pieces, pierce a small hole through both at each eyelet mark, and set an eyelet.
4. Inward chevrons identify every through-page construction. Round circle-and-dot marks elsewhere are electrical test points, not vias.
5. Measure A to B. **Flexed** and **Day 7** work as on T0.

	Day 0	Flexed	Day 7
V0			
V1			
V2			
V3			
V4			

H3 GND · NC

3 · Edge wraps

1. Lay 12 mm tape along the top edge so half overhangs, fold it onto the back down to the dashed line, and burnish it.
2. **One sheet:** measure from the front wrap to the back wrap.
3. **Side join:** overlap two sheets' top corners by 10 mm, wrap on wrap. Measure from one to the other, with and without a binder clip.
4. **Stack:** stack three sheets, top edges level, and clip along the edge. Measure from the top sheet's front wrap to the bottom sheet's back wrap, then again while turning the pages.

One sheet, front to back	
Side join, clipped	
Side join, no clip	
Stack, clipped	
Stack, turning pages	

4 · Stacking shorts

1. On two sheets, lay the front pad and the S1 lane, the back pad on the back, and the wire via.
2. Lay sheet A on sheet B, shifted 15 mm to the left so B's S2 shows. A's back pad now lies on B's front pad.
3. Measure from A's S1 to B's S2, as listed below.

No interleaf	
Pressing hard	
Plain paper between	
Vellum between	

Pass or fail

- **Vias and wraps:** within 1 Ω of V0 and steady; flicker or OL fails.
- **Shorts:** any reading without an interleaf means stacked two-sided pages need one. With an interleaf it must read OL.

Results for the standard

Duplex within 1.5 mm?	
Best via	
Edge rails reliable?	
Interleaf needed?	

H4 BUS B · NC